

Karnaugh Map (K-map):

K-map is an array of square cells in which each square represents a binary value of input variables. The number of square of square in k-map is equal to total no. of possible input variable combination.

For 2 variable no. of square is ($2^2=4$)

		B	
		0	1
A	0	$A'B'$	$A'B$
	1	AB'	AB

For 3 variable no. of square is ($2^3=8$)

		BC			
		00	01	11	10
A	0	m_0	m_1	m_3	m_2
	1	m_4	m_5	m_7	m_6

For 4 variable no. of square is ($2^4=16$)

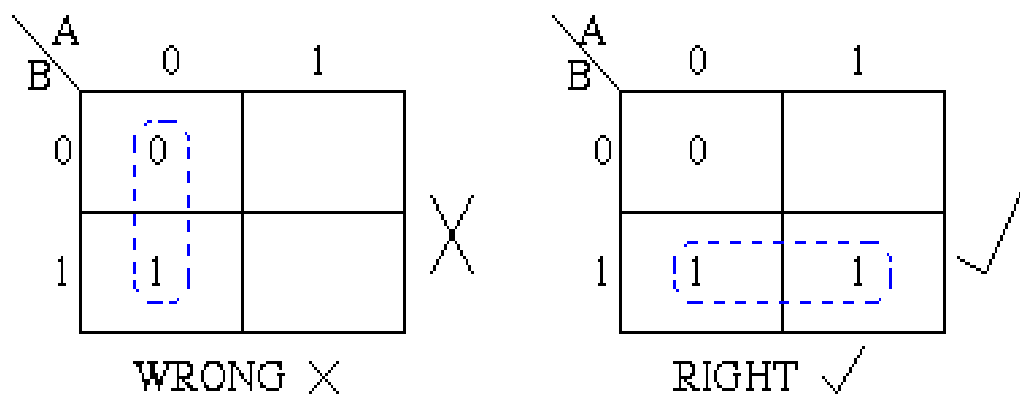
		CD			
		00	01	11	10
AB	00	$A'B'C'D'$	$A'B'C'D$	$A'B'CD$	$A'B'CD'$
	01	$A'BC'D'$	$A'BC'D$	$A'BCD$	$A'BCD'$
	11	$ABC'D'$	$ABC'D$	$ABCD$	$ABCD'$
	10	$AB'C'D'$	$AB'C'D$	$AB'CD$	$AB'CD'$

How to group cells in K-map?

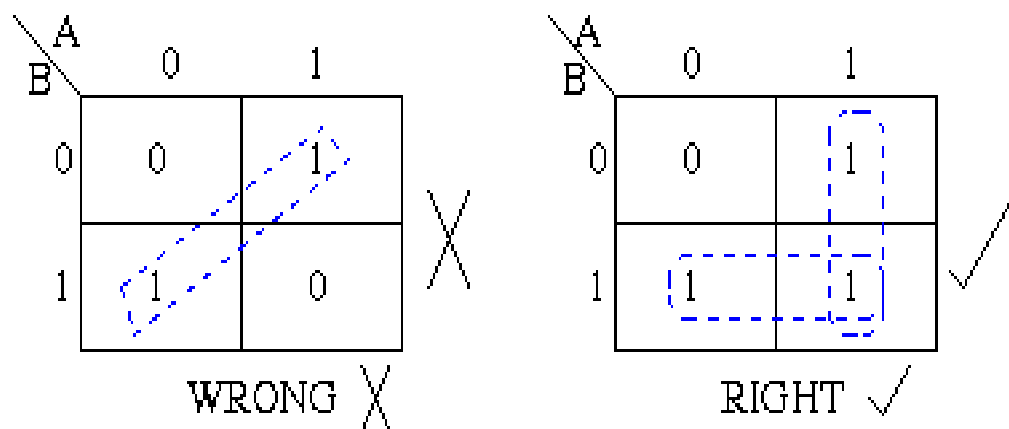
- Groups should contain as many '1' cells (i.e. cells containing a logic 1) as possible and no blank cells.
- Groups can only contain 1, 2, 4, 8, 16 or 32... etc. cells (powers of 2).
- A '1' cell can only be grouped with adjacent '1' cells that are immediately above, below, left or right of that cell; no diagonal grouping.
- Groups of '1' cells can overlap. This helps make smaller groups as large as possible, which is an advantage in finding the simplest solution.
- The top/bottom and left/right edges of the map are considered to be continuous, so larger groups can be made by grouping cells across the top and bottom or left and right edges of the map. (more explained in the picture below)
- There should be as few groups as possible.

Rules for Grouping: (for SOP form) **1's**

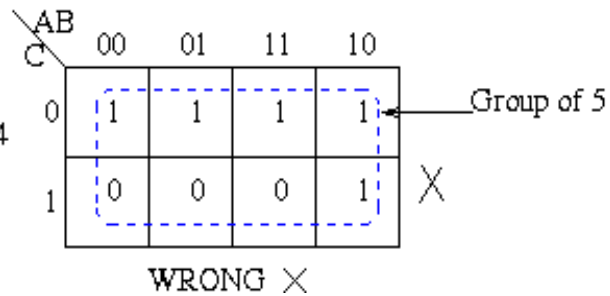
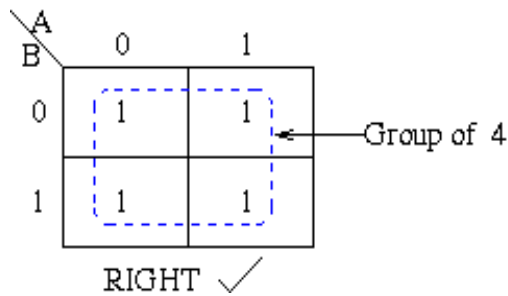
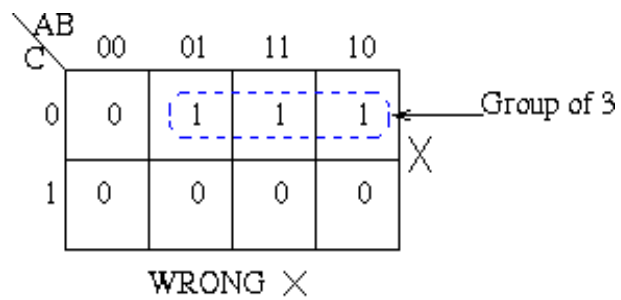
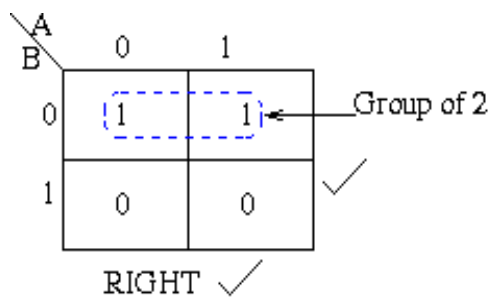
1. Groups may not include any cell containing a zero.



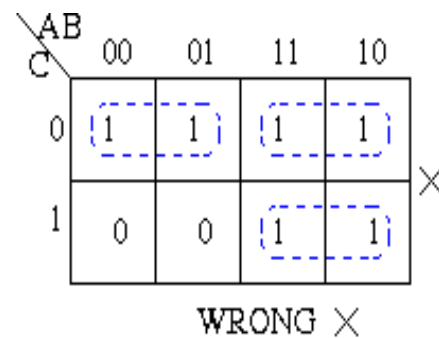
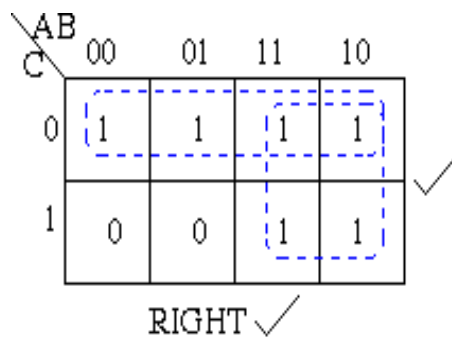
2. Groups may be horizontal or vertical, but not diagonal.



3. Groups must contain 1, 2, 4, 8, or in general 2^n cells. That is if $n = 1$, a group will contain two 1's since $2^1 = 2$. If $n = 2$, a group will contain four 1's since $2^2 = 4$.

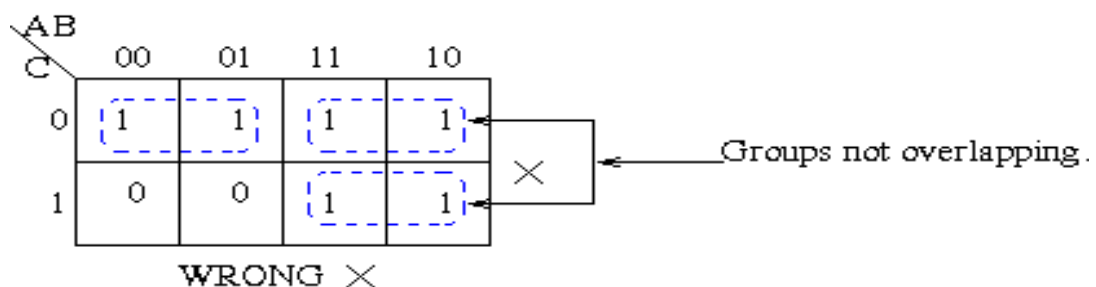
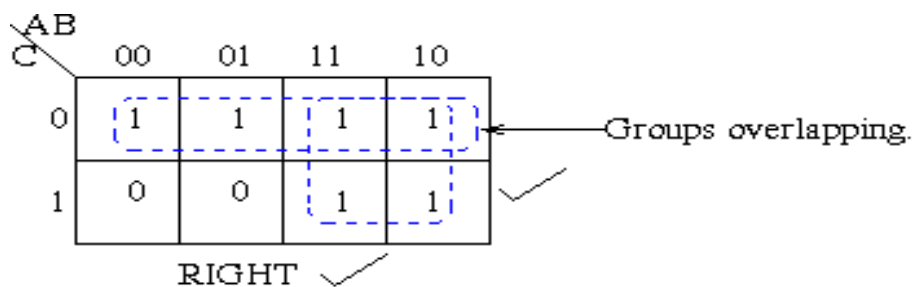


4. Each group should be as large as possible.

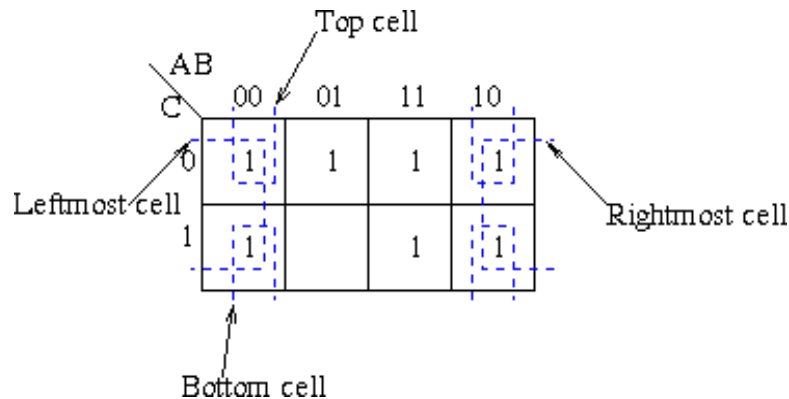


(Note that no Boolean laws broken, but not sufficiently minimal)

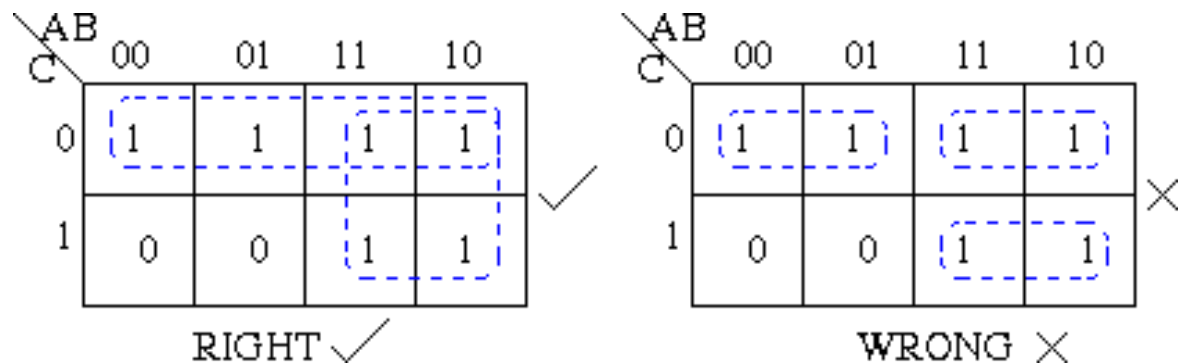
5. Groups may overlap.



6. Groups may wrap around the table. The leftmost cell in a row may be grouped with the rightmost cell and the top cell in a column may be grouped with the bottom cell.



7. There should be as few groups as possible, as long as this does not contradict any of the previous rules.



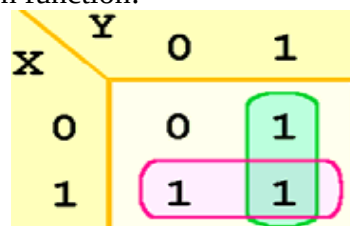
Rules of minimization in K-Map

1. While grouping, you can make groups of 2^n number where $n=0, 1, 2, 3, \dots$
2. You can either make groups of 1's or 0's but not both.
3. Grouping of 1's lead to Sum of Product form and Grouping of 0's lead to Product of Sum form.
4. While grouping, the groups of 1's should not contain any 0 and the group of 0's should not contain any 1.
5. The function output for 0's grouping should be complemented as F' .

Q. Simplify the Boolean function $F = x'y + xy' + xy$ using K-Map.

Solⁿ:

K-map for given function:

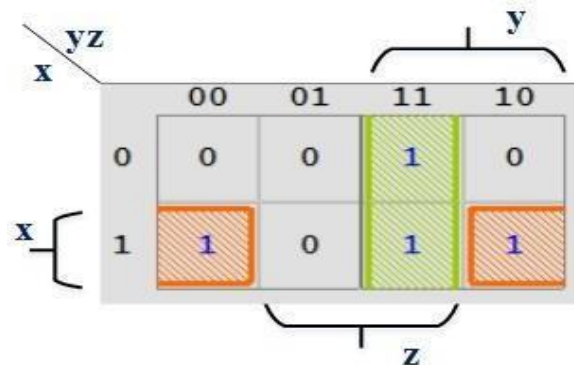


So, after simplification, $F = X + Y$

Q. Simplify the Boolean function, $F = X'YZ + XY'Z' + XYZ + XYZ'$ using K-Map.

Solⁿ:

K-map for given function:



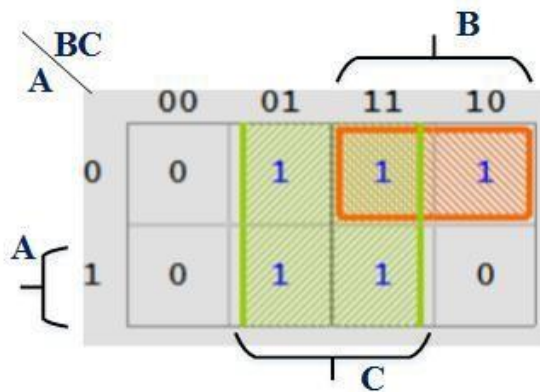
After simplification, $F = YZ + XZ'$

Q. Simplify the Boolean function: $F = A'C + A'B + AB'C + BC$ using K-Map.

Soln:

$$F = A'C + A'B + AB'C + BC = A'BC + A'B'C + A'BC + A'BC' + AB'C + ABC + A'BC$$

K-map for given function:

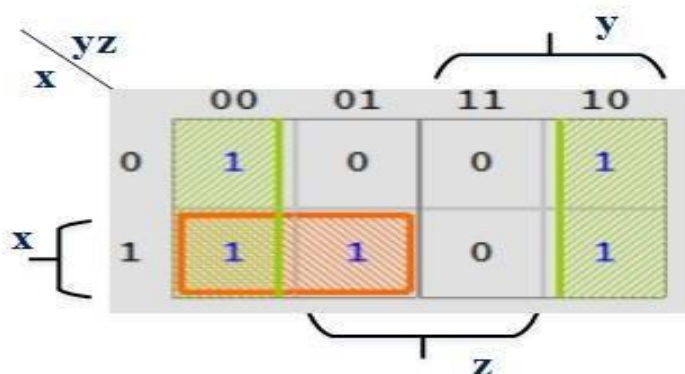


After simplification, $F = C + A'B$

Q. Simplify the Boolean function: $(X, Y, Z) = \sum(0, 2, 4, 5, 6)$ using three variable K-map.

Solⁿ:

K-map for given function:



After simplification, $F = Z' + XY'$

Q. Simplify the Boolean function $F = A'B'C' + B'CD' + A'BCD' + AB'C'$ using K-map.

Solⁿ:

$$\begin{aligned} F &= A'B'C' + B'CD' + A'BCD' + AB'C' \\ &= A'B'C'(D + D') + B'CD(A + A') + A'BCD' + AB'C'(D + D') \\ &= A'B'C'D + A'B'C'D' + AB'CD + A'B'CD + A'BCD' + AB'C'D + AB'C'D' \end{aligned}$$

K-map for given function:

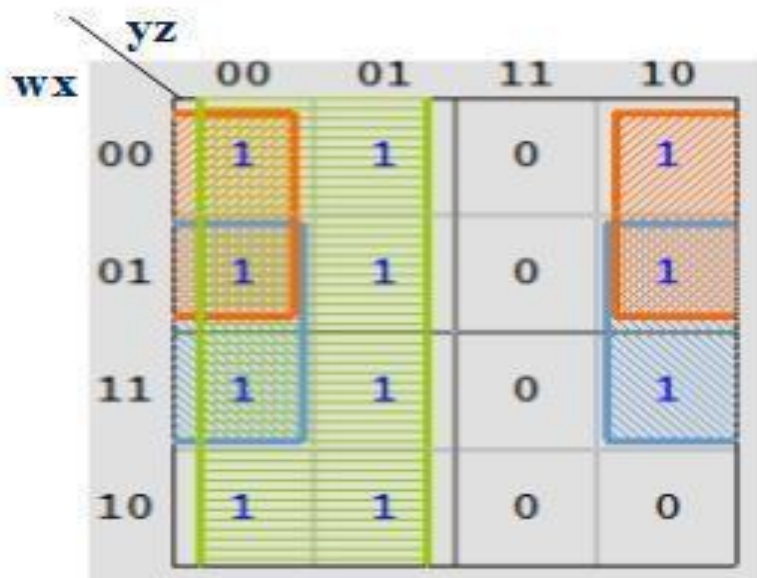


After simplification, $F = B'D' + B'C' + A'CD'$

Q. Simplify the Boolean function $(W, X, Y, Z) = \sum(0, 1, 2, 4, 5, 6, 8, 9, 12, 13, 14)$.

Solⁿ:

K-map for given function:



After simplification, $F = Y' + W'Z' + XZ'$

Product of Sum (POS): 0's

- The given POS expression consists of Maxterms.
- Corresponding to every Maxterm, we enter a '0s' in K-Map and enter '1s' in the remaining cell of K-Map.
- We group 0s for carrying out simplification using grouping techniques of adjacent cells.

Q. Simply the Boolean function $(A, B, C, D) = \prod(0, 1, 2, 3, 4, 10, 11)$ in POS.

Solⁿ:

K-map for given function:

		CD			
		00	01	11	10
AB	00	0	0	0	0
	01	0	1	1	1
	11	1	1	1	1
	10	1	1	0	0

From Map,

$$F' = A'B' + B'C + A'C'D'$$

So,

$$F = (A + B)(B + C)(A + C + D)$$

Q. Simplify the Boolean function: $F = \sum(0, 1, 2, 5, 8, 9, 10)$ in (a) sum of product (SOP) and (b) product of sum (POS).

Solⁿ

a. Sum of products simplification

		CD			
AB		00	01	11	10
	00	1	1	0	1
	01	0	1	0	0
	11	0	0	0	0
	10	1	1	0	1

$$F = B'D' + B'C' + A'C'D$$

b. Product of sums simplification

		CD			
AB		00	01	11	10
	00	1	1	0	1
	01	0	1	0	0
	11	0	0	0	0
	10	1	1	0	1

$$F' = AB + CD + BD' \text{ So, } F = (A' + B')(C' + D')(B' + D)$$

